CROSS-LINGUAL TRANSFER FOR LOW-RESOURCE LANGUAGE MODELING

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Motivation

Adapting Large Language Models (LLMs) to low-resource languages is hindered by tokenizer mismatch and data scarcity. While English-centric models like RoBERTa possess strong linguistic priors, their subword vocabularies are inefficient for morphologically rich languages like Swahili, leading to heavy fragmentation (high fertility) and diluted semantic signals. This study evaluates how different adaptation strategies balance the tradeoff between representation efficiency, computational cost, and knowledge preservation.

Experimental Setup

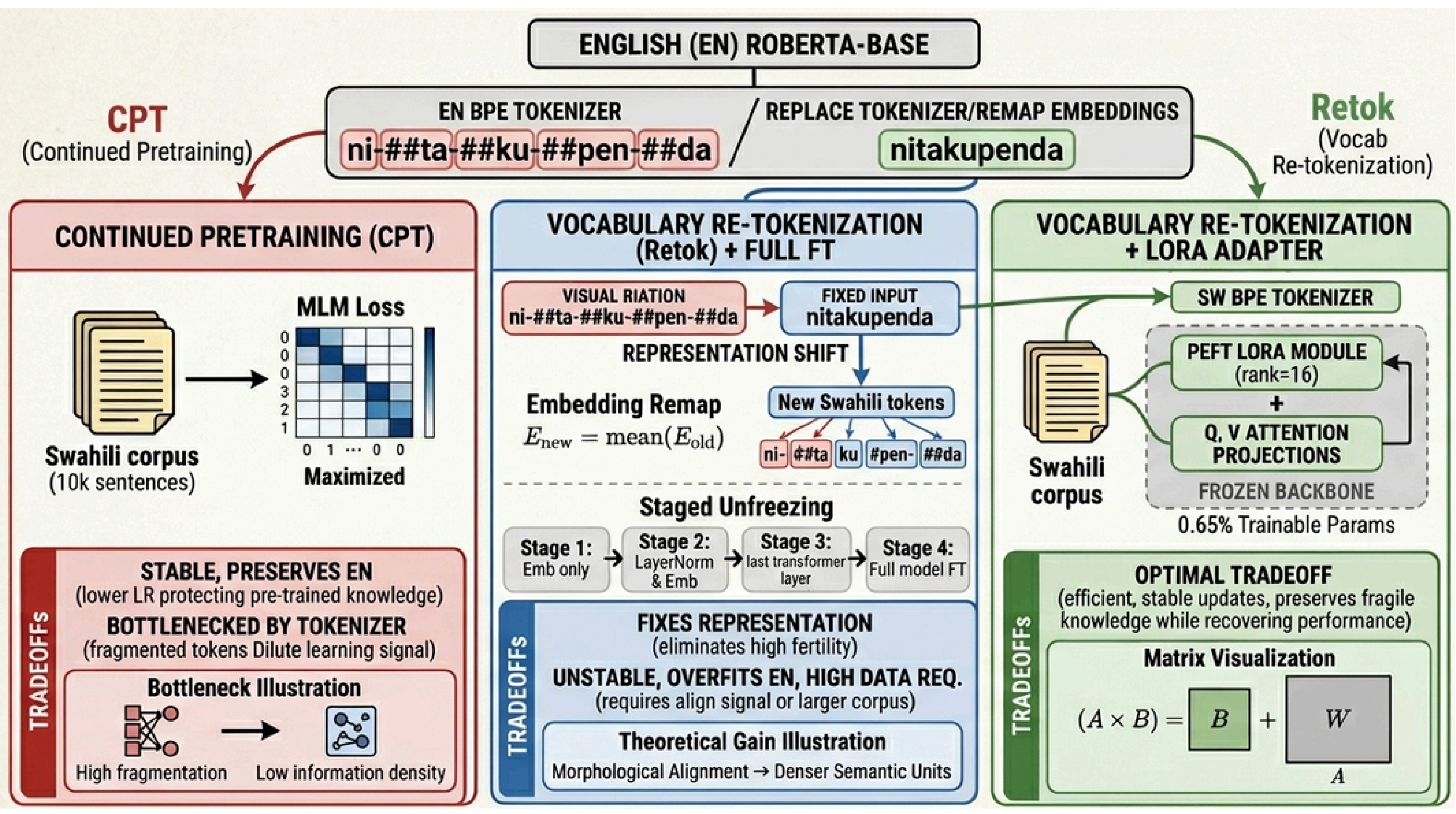

Figure 1. Three strategies for low-resource adaptation

A RoBERTa-base model was pretrained on 100k English sentences (Wikitext-103) before adaptation to a low-resource 10k sentence Swahili corpus. Three strategies are compared: Continued Pretraining (CPT) using the original English tokenizer, Vocabulary Re-tokenization (Retok) with a custom 12k Swahili BPE and 4-stage unfreezing, and LoRA Adaptation (r=16) on the re-tokenized backbone. Swahili embeddings are initialized via deterministic subword mean-pooling:

$$E_{new}(t_{sw}) = \frac{1}{n}\sum_{i=1}^{n} E_{old}(t_{en}^{(i)})$$

Evaluation follows a three parts. (1) Perplexity (PPL = exp(MLM loss)) measures Swahili adaptation vs. English retention on Wikitext-103. (2) Morphological Reconstructio if the model understands intra-word logic or is merely memorizing surface forms. The procedure involves identifying words with > 2 subtokens and masking an internal subtoken—typically a tense or subject marker—to require the model to reconstruct the specific morphological unit using both the surrounding sentence and word-internal context. Success is measured by Top-k prediction accuracy, which effectively isolates the impact of tokenizer granularity on the model's ability to learn morphological patterns. (3) Zero-Shot Transfer on AfriSenti dataset of Swahili tweets categorized into Positive, Neutral, and Negative classes. Evaluation is performed via Log-Sum-Exp scoring:

$$\mathrm{score}(y \mid x) = \log \sum_{w \in \mathcal{V}_y} \exp\left(\sum_{t \in w} \log P(t \mid x)\right)$$

This setup isolates the impact of tokenizer efficiency against parameter-efficient fine-tuning under extreme data constraints.

Results

Strategy	SW PPL	EN PPL	Zero-Shot F1	Params
Base (EN)	2.60*	**6.16**	0.19	0%
Method 1: CPT	13.4*	6.16	0.16	100%
Method 2: Full FT	371	1200	0.17	100%
Method 3: LoRA	**321**	**407**	**0.26**	**0.65%**

Table 1: Performance Trade-offs Across Adaptation Strategies.
***Note**: PPL and Morphological scores for CPT are artificially inflated by extreme token fragmentation; they represent subword-pattern memorization rather than semantic understanding.

Method	Top-3 Predictions	Linguistic Quality
Continued PT	ya, wa, kwa	**Low** (Only high-freq function words)
Retok + FT	na, ya, asubuhi	**Medium** (Mix of grammar and semantics)
Retok + LoRA	**asubuhi, mchana, leo**	**High** (Contextually appropriate nouns)

Table 2: Fill-Mask Comparison on Greeting Context
Prompt: "Habari za [MASK]?" (English: "News of [MASK]?")

- The Tokenizer Efficiency Gap: Replacing the English BPE with a Swahili-specific tokenizer reduced Fertility by 73.4% (dropping from 5.38 to 1.43 tokens/word). This "densification" allows the model to process nearly 4x more semantic content within the same context window.
- Morphological vs. Statistical Learning: Re-tokenized models (Methods 2 & 3) significantly outperformed CPT in intra-word logic. CPT's low PPL was revealed as a statistical artifact of predicting fragmented subwords, whereas LoRA (Method 3) successfully reconstructed masked tense and subject markers in complex verbs.
- Retention and Zero-Shot Transfer: Full Fine-Tuning caused catastrophic forgetting, with English PPL exploding to > 1200. Conversely, LoRA stabilized the model, achieving the highest Zero-Shot F1 (0.26) by mapping Swahili tokens to frozen English semantic priors.
- The "Goldilocks" Strategy: For extreme low-resource scenarios, Retok + LoRA provides the optimal balance—fixing the input representation while preserving the structural intelligence of the pre-trained backbone.

Conclusions

- Tokenizer quality dominates adaptation performance more than optimization strategy.
- MLM alone is insufficient for true semantic grounding in zero-shot tasks; however, LoRA provides the best tradeoff between learning new language distributions and retaining cross-lingual priors.
- For low-resource scenarios, the optimal pipeline is: Retokenization -> Embedding Remapping -> LoRA Adaptation.